

AP Precalculus: 2026–2027

Unit 1: Polynomial and Rational Functions Lesson 1.13: Function Model Selection and Assumption Articulation

Primary Objective: Students will select an appropriate function model (linear, quadratic, cubic, polynomial, or piecewise) for a contextual scenario by analyzing rates of change and shape/context clues, and will articulate the underlying assumptions and domain/range restrictions embedded in the model. *Skills: MP 2.A, MP 3.C*

Priority Ideas & Skills

AP Skills

MP 2.A — Identify information from mathematical representations (tables, graphs, descriptions) to determine function type.

MP 3.C — Support model-selection conclusions with a logical rationale.

Key Understandings

- Constant 1st differences $\Rightarrow$ linear; constant 2nd differences $\Rightarrow$ quadratic; constant n th differences $\Rightarrow$ degree- n polynomial. (EK 1.13.A.1–A.5)
- Area/2-D context $\Rightarrow$ quadratic; volume/3-D context $\Rightarrow$ cubic. (EK 1.13.A.3)
- A degree- n polynomial can pass through $n+1$ points with distinct inputs. (EK 1.13.A.6)
- Piecewise models arise when behavior changes over different intervals. (EK 1.13.A.7)
- Every model carries assumptions about consistency, co-variation, and restricted domain/range. (EK 1.13.B.1–B.4)

Vocabulary, Concepts, & Theorems

first differences

The differences between consecutive output values of equally-spaced inputs:
 $\Delta y_i = y_{i+1} - y_i$.

second differences

Differences of the first differences; constant for quadratic models.

n th differences

Differences taken n times; constant and nonzero for degree- n polynomial models.

domain restriction

A limit on the input values that is mathematically, contextually, or practically required.

range restriction

A limit on the output values (e.g., integer-only outputs due to rounding).

piecewise-defined function

A function defined by different formulas over different intervals of its domain.

model assumption

A simplifying claim about the real world that must hold for the model to be valid.

Activate Prior Knowledge & Spiral Review

Spiral review (warm-up) covers: computing first and second differences of a data set; deciding whether a rate of change is constant (linear) or changing (nonlinear). These skills connect directly to the difference-table decision tree introduced today.

Hook

“A scientist hands you a table of mystery data. How do you decide which function type to use?”

Display a short table with six equally-spaced inputs and outputs. Ask cold-call: “Is this linear, quadratic, or something else? How can you tell without graphing it?” Let students debate for 90 seconds before revealing the difference-table approach.

Lesson

Part 1 (12 min): Difference-Table Decision Tree.

Given a table with equally-spaced inputs, compute differences:

1. Compute 1st differences (Δy).
2. If constant $\Rightarrow$ **linear** (degree 1).
3. If not constant, compute 2nd differences ($\Delta^2 y$).
4. If constant $\Rightarrow$ **quadratic** (degree 2).
5. Continue: constant n th differences $\Rightarrow$ degree- n polynomial.

Pose: “If 1st differences grow linearly themselves, what does that tell you?” (Answer: 2nd differences are constant, so the data is quadratic.)

Part 2 (10 min): Shape & Context Clues.

When no table is given, use context:

- Roughly constant rate of change $\Rightarrow$ linear.
- Area or 2-D measurement $\Rightarrow$ quadratic.
- Volume or 3-D measurement $\Rightarrow$ cubic.
- Multiple local maxima/minima or multiple real zeros $\Rightarrow$ higher-degree polynomial.
- Behavior changes abruptly at a threshold $\Rightarrow$ piecewise.

Part 3 (10 min): Stating Assumptions (LO 1.13.B).

Every model rests on assumptions. Template: “This model assumes that [quantity/behavior] remains [consistent/proportional/valid] over [domain interval]. The domain is restricted to [values] because [mathematical / contextual reason].”

Walk through a fencing example: “We model the enclosed area as a quadratic in side length. We assume the fencing forms a perfect rectangle and that side length is between 0 and half the total fencing available.”

Active Monitoring

- After Part 1: cold-call “If 3rd differences are constant, what degree polynomial models the data?”
- After Part 2: cold-call “Why can’t you model negative time in the fencing problem?” (domain restriction)
- Watch for students who compute differences but stop at 1st differences even when they are not constant.
- Watch for students who confuse *contextual* domain restrictions (e.g., length > 0) with *mathematical* ones (e.g., denominator $\neq 0$).
- After Part 3: cold-call “What would constant second differences tell you?” — expect “the data is quadratic” or “the rate of change is linear.”

Group Work & Differentiation

- **Tier R (Remediate):** Given four scenarios with model type choices; circle the best model and complete a fill-in assumption sentence.
- **Tier A (Apply):** Compute difference tables for provided data, determine degree, name the model type, and state one assumption with domain restriction.
- **Tier E (Extend):** Write a piecewise model description (two pieces, each with appropriate function type) for a real-world scenario and articulate two assumptions with justification.

Individual Work & Assessment

Exit ticket (3 questions, 1 page): (1) complete a first/second differences table and name the model type; (2) read a scenario description and circle the best model with one reason; (3) state one domain restriction and one assumption for a fencing model. Collect and use to group students before Lesson 1.14.

Reinforcement & Extension

Homework: 5–6 problems covering difference tables, context-to-model matching, assumption articulation, and piecewise model description. Extension: given 4 data points, write the unique cubic model in descriptive form and state what real-world meaning the degree implies. Preview of Lesson 1.14: students will *construct* function models from scratch — today’s selection and assumption skills are the prerequisite.